

Active Women, Social Physique Anxiety, and Exercise

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The purpose of this investigation was to replicate and extend Crawford and Eklund's (1994) investigation of social physique anxiety (SPA) and exercise. Women ($N = 94$) enrolled in physical education activity or major classes participated in the investigation. Data were collected on SPA, weight satisfaction, percent body fat, reasons for exercise, exercise behaviors and preferences, and attitudes toward two aerobic class video presentations featuring a manipulation of physique salience. Consistent with the previous investigation, self-presentational reasons for exercise (body tone, weight control, and physical attractiveness) were positively associated with SPA in both simple correlations and hierarchical analyses controlling for body composition. In contrast to previous findings, SPA was not associated with favorability of attitudes toward either of the video presentations. The inability to fully replicate Crawford and Eklund's (1993) findings raised interesting questions with regard to variables that may moderate or mediate self-presentational anxiety in exercise settings.

Key words: self-presentation, body composition, aerobic dance, attitudes

Recent commentary and research has suggested that self-presentational perspectives may be relevant for understanding exercise behaviors (Crawford & Eklund, 1994; Hart, Leary, & Rejeski, 1989; Leary, 1992; Spink, 1992). The basic premise has been that people engage in selective presentation and omission of aspects of the self to create desired impressions and to avoid undesired impressions with specific people in a particular social encounter (Leary, 1992). Because exercise participation often occurs in social contexts, it seems reasonable that self-presentational strategies would have an impact on exercise behaviors.

There are both acquisitive and protective aspects involved in self-presentation (Arkin, 1981). People are motivated to garner social approval by employing acquisitive strategies in their self-presentations to make desired impressions. Conversely, individuals are also motivated to avoid significant losses in social

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approval or to avoid social disapproval and may use protective strategies in their self-presentations by assuming shielding or unobtrusive postures, or even by abstaining from situations where they believe they are not capable of making a desired impression (Arkin, 1981).

There are differences among people in the degree to which self-presentational exercise motives are important (Klesges, Mizes, & Klesges, 1987; Silberstein, Mishkind, Striegel-Moore, Timko, & Rodin, 1989; Silberstein, Striegel-Moore, Timko, & Rodin, 1988). Nonetheless, acquisitive self-presentation can be intuitively understood as an important process when desires to improve or maintain physical appearance through exercise or to construct a certain social identity or image (such as being fit or athletic) are considered (Leary, 1992). However, physique-related perceptions may also stimulate protective self-presentational behaviors or deter individuals from being active when they have concerns about being negatively evaluated while exercising (Hart et al., 1989; Leary, 1992; Spink, 1992). For example, Bain, Wilson, and Chaikind (1989) reported that the most influential factor limiting the willingness of overweight female exercisers from exercising in public or from attending exercise studios was a concern about being observed and evaluated by others.

Social anxiety is an affective consequence that may be experienced when people doubt their ability to make desired impressions on others (Leary, 1983; Schlenker & Leary, 1982). Because the degree of chronic social anxiety has been associated with a number of protective self-presentational behaviors outside of exercise settings (Dykman & Reis, 1979; McGovern, 1976; Watson & Friend, 1969), it seems likely that similar associations would be found within social exercise settings. Hart et al. (1989) advanced the construct of *social physique anxiety* (SPA) as social anxiety specific to the physique. Theoretically, SPA should be a particularly salient form of social anxiety in social exercise activity because the body is central in such settings, and thus, may be a significant factor influencing exercise behaviors.

Indeed, recent research suggests that SPA may be associated with exercise behavior patterns. Crawford and Eklund (1994) found that, although the frequency and duration of exercise were not associated with SPA, the favorability of attitudes towards exercise settings emphasizing or de-emphasizing the physique salience among college women was significantly associated with SPA. Specifically, SPA responses were negatively associated with the favorability of attitudes toward a video presentation of an aerobic dance setting featuring exercisers dressed in typical aerobic dance fashion attire. Conversely, SPA was positively associated with the favorability of attitudes toward the same setting when the participants were attired in less revealing shorts and T-shirts. Furthermore, consistent with Arkin's (1981) notion of protective self-presentation, SPA was most strongly associated with the favorability of attitudes toward the exercise setting emphasizing the physique. Crawford and Eklund (1994) also found SPA to be positively associated with the emphasis placed upon self-presentational reasons for exercise among the college women.

Hart and Gill (1993) have also found some suggestive results in examining gender, aerobic dance experience, SPA, and aerobic dance exercise behaviors and preferences in a field investigation at a large private fitness facility. Specifically, among the 63 subjects (44 female, 19 male; mean age = 39.4) participating in the investigation, inexperienced exercisers reported greater anxiety specific to

the physique in social exercise settings than did experienced exercisers. Further, and comparatively speaking, women and more experienced exercisers had a greater preference for tighter fitting exercise clothing, whereas men had a greater preference for loose-fitting clothing. Men also were found to have a greater preference for coed aerobic dance classes and a greater awareness of the opposite sex during aerobic dance classes than women had. Finally, significant differences between experienced and inexperienced exercisers were noted in most and least liked floor positions in the aerobic dance class.

In initial validation studies, Hart et al. (1989) found significant physical differences between women scoring relatively high in SPA and those scoring relatively low. Specifically, high-SPA women were significantly heavier, taller, and higher in percentage body fat than low-SPA women. Further, in evaluations by researchers blind to SPA levels, high-SPA women were rated as significantly larger than low-SPA women. Clearly there is a need to examine whether self-presentational anxieties related to the physique contribute to understandings of exercise behavior beyond what might be available via relatively objective measures of body composition. Therefore, the primary purpose of the present investigation was to replicate and extend Crawford and Eklund's (1994) exploration of associations between SPA and exercise by taking body composition into account in the analyses. The second purpose of this investigation was to examine associations between SPA and questions relevant to self-presentation regarding exercise behaviors and preferences used by Hart and Gill (1993) in making comparisons between male and female exercisers and between more and less experienced exercisers.

It was hypothesized that associations reported by Crawford and Eklund (1994) between SPA and favorability of attitudes toward exercise and between SPA and reasons for exercise would be replicated in this investigation even while controlling for body composition. Further, it was hypothesized that SPA would be associated with patterns of response to questions relevant to self-presentation regarding exercise behaviors and preferences.

Method

Subjects

College-age females ($n = 94$) volunteered to participate in this study. The subjects ranged in age from 18 to 23 years ($M = 20.3$), had a mean weight of 136.3 lb ($SD = 19.3$) and a mean height of 66.0 inches ($SD = 2.8$). Mean percent body fat among these women was 21.67 ($SD = 3.96$). Ninety-four percent of the women indicated that they enjoyed exercising, and they reported an average of 6.5 hours per week ($SD = 4.9$) of exercise or physical recreation.

As in Crawford and Eklund's (1994) study, this investigation was limited to women. The decision to focus upon women was based on several observations. First, women make up the vast majority of participants in aerobic dance classes (the exercise setting of focus). Second, evidence suggests that, in comparison to men, women are particularly vulnerable to body dissatisfaction (Rodin, 1992; Silberstein et al., 1988). Third, body-image dissatisfaction and the direction of this dissatisfaction differs between men and women (Cash, Winstead, & Janda,

1986; Fallon & Rozen, 1985; Franzoi & Shields, 1984; Miller, Linke, & Linke, 1980; Silberstein et al., 1988). Finally, display of the physique is a prominent aspect in popular media portrayals of exercise activity for women.

Instruments

Social Physique Anxiety Scale (SPAS). The SPAS is a 12-item self-report inventory developed by Hart et al. (1989) to measure the construct of social physique anxiety. Subjects are asked to indicate the degree to which statements are characteristic or true of them on a 5-point Likert-type scale with anchors of *not at all* (1), *slightly* (2), *moderately* (3), *very* (4), and *extremely* (5). Evidence has been presented to suggest that the SPAS demonstrates construct validity (Hart et al., 1989; McAuley & Burman, 1993), test-retest reliability, internal consistency ($\alpha = .90$), and minimal social desirability bias (Hart et al., 1989). In this investigation, the SPAS was found to have satisfactory internal consistency with an alpha coefficient of .93 (item-to-total correlations ranged from .56 to .79).

Previously, SPAS Item 2 ("I would never worry about wearing clothes that might make me look too thin or overweight") has been found to be problematic (Crawford & Eklund, 1994; Lantz, Hardy, & Ainsworth, 1991; McAuley & Burman, 1993). Crawford and Eklund (1994) suggested that the negative wording of this item creates some confusion in responding (thus resulting in the low item-total correlation) and that rephrasing the item to a positive statement might alleviate this problem. Hence, this item was rephrased to read, "I would worry about wearing clothes that might make me look too thin or overweight." The rephrasing of this item appears to have resolved the problem, with a resulting item-total correlation of .72.

Percent Body Fat (BF%). Skinfolds at four body sites (triceps, abdomen, suprailium, and thigh) were measured by a trained female laboratory assistant using Lange calipers. Body density was calculated via three different equations appropriate to the population. Specifically, Sloane, Burt, and Blyth's (1962) two-site (suprailium and triceps) equation for estimating body density for young women and two three-site generalized equations (triceps, suprailium, and thigh; triceps, suprailium and abdomen) by Pollock, Schmidt, and Jackson (1980) were used. Percentage of fat was calculated from body density via the equation of Brozek, Grande, Anderson, and Keys (1963). Correlations among the estimates of BF% ranged from .90 to .97, and average BF% ranged from 21.5 to 22.5. Given the substantial agreement, the average of the three estimates was used in all statistical analyses.

Weight Satisfaction (WS). WS was determined by asking subjects to identify their current and ideal weights in pounds. The difference between the current and ideal weights was divided by current weight to give a percentage discrepancy from ideal, which was utilized to represent WS. The raw WS score was utilized descriptively, but the absolute value of the discrepancy was used in further analyses. Only three women indicated an ideal weight that was heavier than their current weight, and for the purposes of this investigation, the direction of the dissatisfaction was not considered to be as important as the magnitude of dissatisfaction. As a cautionary note, it should be mentioned that this variable has not gone without criticism (P. Crocker, personal communication, June 4, 1993).

Specifically, even though current/ideal weight discrepancies have been previously used as an index of weight satisfaction (or dissatisfaction) (e.g., Crawford & Eklund, 1994; Silberstein et al., 1988), the magnitude of the discrepancy does not necessarily indicate the degree of satisfaction (or dissatisfaction) with current weight.

Reasons for Exercise Inventory (REI). The REI is a 24-item self-report inventory with seven subscales purporting to address motives for engaging in exercise activities (Silberstein et al., 1988). The seven REI subscales include exercising for Weight Control, for Fitness, for Health, to improve Body Tone, to improve overall Physical Attractiveness, for Mood Enhancement, and for Enjoyment. Silberstein et al. (1988) reported that the seven subscales demonstrated adequate internal consistency; however, Crawford and Eklund (1994) found it necessary to delete items from the Weight Control, Mood Enhancement, Physical Attractiveness, and Enjoyment subscales to obtain satisfactory alpha coefficients (Nunnally, 1978). Subscale scores in this investigation were calculated using the identical items used in the Crawford and Eklund (1994) investigation because the purpose of the study was to replicate and extend the previous investigation. The REI subscales all demonstrated sufficient internal consistency with all appropriate item-total correlations in excess of .44.

Exercise Behaviors and Preferences. Twelve questions regarding exercise behaviors and preferences previously piloted and employed by Hart and Gill (1993) were presented to subjects. Hart and Gill's (1993) questions were developed to examine self-presentation issues in aerobic dance exercise settings such as personal appearance, social evaluation, sociability, and exercise atmosphere of the class. The 5-point Likert-type response format includes appropriate-for-question anchors such as *definitely prefer not to*, *not at all*, *very negatively*, *very unimportant* for values of 1 and *definitely prefer*, *all the time*, *very positively*, and *very important* for values of 5. These questions do not constitute a scale per se, and hence, analyses were conducted on individual items.

Stimulus Materials. Two videotapes of aerobic exercise classes (each 2 minutes in duration) professionally developed for the Crawford and Eklund (1994) investigation served as stimulus materials. Except for the manipulation, these videos were identical in all regards, including participants, relative floor positions, music, the simple aerobic dance routine, camera angles, and timing of camera angle changes. The portrayal of the aerobics classes was designed to be consistent with classes that might be encountered at any local health club. A total of 9 exercisers were visible in the videotaped exercise classes. Among these were a female instructor and 7 female and 1 male participants. As judged by independent viewers of the videos, the class members all appeared to be between the ages of 20 and 35, reasonably fit, proportional in body composition, coordinated, and experienced in aerobics.

The manipulation of the aerobic dance setting involved changing only the attire that the participants were wearing. In the aerobics fashion attire video, participants were wearing tights and thong leotards that emphasized the physique/figure. This attire was not risqué or unusual, but nonetheless, it was relatively revealing with regard to the physique. In the shorts and T-shirts attire video, participants wore contemporary-style T-shirt and shorts workout gear over their tights and leotards to de-emphasize the salience of the physique/figure.

Crawford and Eklund (1994) had selected aerobic dance as an exercise

activity for the stimulus materials because it is popular, and frequently practiced in group settings in which self-presentation, particularly with regard to the physique, may be a central factor. Further, there is wide variety in acceptable workout attire evident across aerobic dance classes. Standard attire in some classes emphasizes the body, whereas other classes feature a de facto dress code that is much less revealing. These notable differences allowed for the manipulation of physique salience in the investigation stimulus materials that was consistent with what can be found in natural settings.

Attitudes Toward Exercise Settings (ATES). Five items used in the Crawford and Eklund (1994) investigation were used to assess subject attitudes toward the exercise settings. These items were scored on a 5-point Likert scale, with anchors of *not at all* (1) and *extremely* (5). The questions regarded subjects' attitudes toward the favorability of participation in the exercise setting, including "This would be a good group to exercise with," "I would be comfortable exercising with this class," "I would enjoy exercising with this group of people," "I would feel out of place in this exercise class," and "I would like participating with this class regularly." The ATES was completed for both the aerobics fashion attire video presentation (ATES-aerobics fashion attire) and the shorts and T-shirt attire video presentation (ATES-shorts and T-shirt attire). From these ratings, a comparative favorability score (ATES-comparative favorability) of the two exercise settings was also calculated by subtracting ATES-aerobics fashion attire ratings from ATES-shorts and T-shirt attire ratings. The alpha coefficient of the ATES-shorts and T-shirt attire responses in this investigation was .88 (item-total correlations from .44 to .86), and the alpha coefficient of the ATES-aerobics fashion attire responses was .85 (item-total correlations from .52 to .80).

Procedures

Female subjects were recruited from undergraduate physical education major classes and physical activity classes. Women interested in taking part were given appointments for data collection. Informed consent procedures were completed upon arrival for the appointment. Instructions designed to minimize social desirability effects were given, and data were collected on demographics, SPAS, REI, and WS.

Upon collection of these data, subjects viewed one exercise setting videotape and responded to the appropriate ATES questionnaire. Subsequently, subjects viewed the other exercise setting videotape and responded to the second ATES questionnaire. The presentation of videos was balanced across subjects to manage order effects, although statistical examination of data revealed no significant order effects in the ATES-aerobics fashion attire ratings, $t(92) = .29, p > .05$, nor in the ATES-shorts and T-shirt attire ratings, $t(92) = .94, p > .05$. After completing the ATES questionnaires, subjects responded to the 12 Exercise Behaviors and Preferences questions. Upon completion of inventory data collection, skinfold measurement procedures were conducted in a room providing suitable privacy.

Results

Descriptive statistics for SPA, WS, BF%, ATES, and REI are presented in Table 1. Particularly notable was that the subjects had a modest preference for the

Table 1 Descriptive Statistics on SPA, WS, BF%, ATES, and REI

Variable	<i>M</i>	<i>SD</i>	Range	Alpha
SPA	37.31	9.81	13–57	0.93
WS	0.06	0.05	–.05–.26	—
BF%	21.67	3.96	15.1–33	—
ATES				
Shorts and T-shirt attire	18.93	4.00	7–25	0.88
Aerobics fashion attire	16.74	4.30	5–25	0.86
Comparative favorability	2.19	4.24	–10–17	—
REI				
Weight control	11.13	2.19	2–14	0.76
Fitness	21.85	3.84	10–28	0.74
Mood enhancement	12.98	4.27	3–21	0.81
Health	22.21	4.46	10–28	0.84
Physical attractiveness	9.34	3.40	2–14	0.78
Enjoyment	6.14	3.17	2–14	0.81
Body tone	15.94	4.01	6–21	0.80

Note. SPA = social physique anxiety; WS = weight satisfaction; BF% = percent body fat; ATES = attitudes toward exercise settings; REI = Reasons for Exercise Inventory.

shorts and T-shirt attire video, mean BF% was about 22%, and they perceived their idea body weight to be about 6% less than their current weight. Also, Table 2 presents the correlations among of SPA, BF%, ATES variables, and REI subscales. A minimum magnitude has been identified for statistical significance ($r > .20$) at $p < .05$; however, it should be recalled that margin for emergence of Type I errors is reasonably large in considering a correlation matrix of this size.

Prediction of Attitudes Toward Exercise Settings

Pearson correlation coefficients were calculated to examine the relationships between SPA, WS, BF%, and ATES ratings. Correlations between ATES (aerobics fashion attire, shorts and T-shirt, and comparative favorability) and SPA, BF%, and WS ranged from $-.07$ to $.20$, and none were significant.

A repeated measures multiple regression analysis (see Michela, 1990; Pedhazur, 1982) was conducted to further explore hypothesized associations between ATES and predictor variables while controlling for body composition. Specifically, the dependent variable for the analysis was ATES ratings for the videos (both aerobics fashion attire and shorts and T-shirt ratings). The order of entry of predictor variables and rationale for each step in the hierarchical analysis were as follows:

1. The between-subject factors of BF%, SPA, and WS were entered as a block in the first step to account for individual differences in these variables. No preferential order of entry was specified because none were significantly correlated with the criterion variable.

Table 2 Pearson Product Moment Correlations Among WS, BF%, ATES, and REI Variables

				ATES			REI						
	SPA	WS	BF%	1	2	3	4	5	6	7	8	9	
WS		.42	1.00										
BF%		.36	.72	1.00									
ATES													
1. AFA		-.11	-.07	-.10	1.00								
2. STSA		.07	.02	.11	.48	1.00							
3. CF		.17	.09	.20	-.55	.46	1.00						
REI													
4. Weight control		.57	.44	.19	-.02	.11	.12	1.00					
5. Fitness		-.18	.02	-.01	.29	-.04	-.33	-.06	1.00				
6. Mood		.08	-.06	-.07	.27	.10	-.18	.04	.39	1.00			
7. Health		-.29	-.01	-.05	.32	.11	-.21	-.18	.74	.37	1.00		
8. Attractiveness		.26	.10	.04	.20	.23	.01	.48	.07	.16	-.05	1.00	
9. Body tone		.30	.15	.05	.16	.13	-.04	.50	.31	.18	.11	.53	1.00
10. Enjoyment		-.05	-.19	-.20	.32	.18	-.16	.19	.20	.36	.15	.34	.32

Note. WS = weight satisfaction; BF% = percent body fat; ATES = attitudes toward exercise settings; AFA = aerobic fashion attire; STSA = shorts and T-shirt attire; CF = comparative favorability; REI = Reasons for Exercise Inventory; *rs* > .20 are statistically significant at *p* < .05.

2. Ninety-three dummy-coded subject vectors (*N* = 1) were entered at the second step to uniquely identify each subject and to remove the variance attributable to subjects (Michela, 1990; Pedhazur, 1982). The entry of variables in the first two steps accounts for both known and unexplained subject variability and hence allows the remaining variables to be tested for effects independent of between-subject effects.
3. The dummy-coded video variable (shorts and T-shirt attire = 1; aerobics fashion attire = 0) was entered on the third step to test for a main effect associated with video presentation.
4. The interactions between video and BF%, SPA, and WS were entered sequentially in the last three steps to allow examination of association between video presentation and the relevant variables. The order of entry in these steps were a consequence of the desire to first control for body composition before examining SPA and WS contributions in the order of importance reported by Crawford and Eklund (1994).

Significance tests for between-subject effects used the residual between-subject mean square as the error term, and within-subject effects were tested with the appropriate residual within-subject mean square as the error term.

As can be observed in Table 2, intercorrelations among the predictor variables revealed a potential multicollinearity problem, with WS being correlated

Table 3 Repeated-Measures Multiple Regression Results Predicting Attitudes Toward Exercise Settings

	R^2 change	Unstandardized final regression coefficients
Step 1		
WS	.001	-39.44
SPA		-0.09
BF%		0.37
Step 2		
Subject vectors	.692*	—
Step 3		
Video	.065*	-5.50
Step 4		
BF% $\times$ Video	.010	0.29
Step 5		
SPA $\times$ Video	.003	0.06
Step 6		
WS $\times$ Video	.003	-13.92
Constant		14.63
R^2 model	.774	

Note. WS = weight satisfaction; SPA = social physique anxiety; BF% = percent body fat; Video = dummy-coding for video presentation (1 = shorts & T-shirt attire; 0 = aerobics fashion attire).

* $p < .001$.

with BF% at .72. However, sufficient tolerance (Tabachnick & Fidell, 1989) was found among BF%, SPA, and WS to reasonably include all in the regression analysis.

The only significant R^2 increments in the hierarchical analysis were observed in the entry of the dummy-coded subject vectors and in the subsequent entry of the dummy-coded video variable (see Table 3). These results indicate that subject variability unexplained by BF%, SPA, and WS played the largest role in accounting for variability in ATEs. The significant incremental R^2 associated with the entry of the video variable indicates a within-subjects video preference. The positive weighting of the video regression coefficient on that step ($b = 2.1862$) reveals the preference to be in favor of shorts and T-shirt attire setting. However, none of the final model unstandardized regression coefficients identified (BF%, SPA, WS, and video nor interactions between video and BF%, SPA, and WS were significant.

REI and SPA

Both correlation coefficients and hierarchical multiple regression were used to examine the hypothesis that SPA would be associated with reasons for taking

part in exercise. Four of the seven REI subscales were significantly correlated with SPA. Specifically, SPA was significantly correlated with self-presentational reasons for exercise (those related to the development or maintenance of easily observable physical qualities) including exercising for Body Tone ($r = .30, p < .004$), for Weight Control ($r = .57, p < .001$), and for improving Physical Attractiveness ($r = .26, p < .014$). Reasons for exercise largely independent of self-presentation were not significantly correlated with SPA (Fitness $r = -.18$; Mood Enhancement $r = .08$; Enjoyment $r = -.05$) with the exception of exercising for Health ($r = -.29, p < .005$). As might be guessed upon examination of correlations between REI subscales and BF% in Table 4, partial correlations between SPA and REI subscales controlled for BF% were virtually identical to corresponding simple correlations.

Hierarchical multiple regression analyses were conducted to assess the hypothesized association between REI subscales as predictor variables and the criterion variable SPA while controlling for body composition. Prior to conducting these analyses, intercorrelations among REI subscale predictors were examined, and interesting patterns emerged. These intercorrelations ranged from $-.18$ to $.53$, with the exception of a $.74$ correlation between Health and Fitness (see Table 2). Interestingly, all of the self-presentational reasons for exercise are significantly and positively correlated with one another (ranging from $.48$ to $.53$). Similarly, with the exceptions of correlations of Enjoyment with Health ($r = .15$) and Fitness ($r = .20$), correlations among nonpresentational reasons were also significantly

Table 4 Hierarchical Regression Analysis Examining Relationships Between SPA and REI Controlled for Body Composition

	Prediction of SPA				
	Beta	Semipartial r	R^2	Adj R^2	R^2 change
Step 1					
Percent body fat	0.36*		0.13	0.12	
Step 2					
Self-presentational reasons for exercise			0.39	0.36	0.26
Weight control	0.50*	.41*			
Physical attractiveness	−0.01	−.01			
Tone	0.04	.03			
Step 3					
Nonpresentational reasons for exercise			0.48	0.43	0.09
Fitness	−0.10	−.06			
Mood	0.23*	.20*			
Health	−0.21	−.14			
Enjoyment	−0.16	−.13			

Note. Step 1, $F(1, 92) = 13.51, p < .001$; Step 2, $F(4, 89) = 13.99, p < .001$; Step 3, $F(8, 85) = 9.69, p < .001$.

* $p < .05$.

and positively intercorrelated (ranging from .36 to .74). However, the majority of correlations between presentational and nonpresentational reasons of exercise were not significant, with significant correlations observed only between Tone and Fitness ($r = .31$), Tone and Enjoyment ($r = .32$), and Enjoyment and Attractiveness ($r = .34$). Further examination of this predictor set to allay potential multicollinearity concerns revealed sufficient tolerance (Tabachnick & Fidell, 1989) to reasonably include all in the regression analysis.

As can be seen in Table 4, BF% was found to be a significant predictor of SPA ($p < .001$) in Step 1 and accounted for about 13% of the variance. Self-presentational reasons for exercise were entered on Step 2 and were found to contribute significantly to the prediction of SPA ($p < .001$) although Weight Control ($p < .001$) was the only variable to make a significant contribution. The addition of the self-presentational reasons for exercise increased the variance accounted for by about 26 percentage points.

The addition of the nonpresentational reasons for exercise in Step 3 also significantly improved the prediction of SPA ($p < .001$) by accounting for an additional 9 percentage points of variance. Among these variables, Mood was the only reason for exercise to make a significant contribution ($p < .02$), although the entry of this variable, particularly with a positive β , is somewhat puzzling. Specifically, in simple correlations, Mood is uncorrelated with SPA and is significantly and positively correlated with Health, Fitness, and Enjoyment. Health and Fitness are both significantly and negatively correlated in multiple partial correlations with SPA prior to Step 3, whereas multiple partial correlation of Mood to SPA is nonsignificant. Given these circumstances, and the observation that the β s for Health, Fitness, and Enjoyment are all negative (although not significant), it seems likely that the entry of the nonpresentational reasons for exercise as a group increases the variance accounted for in the model via suppression.

SPA and Exercise Behaviors and Preferences

Table 5 displays the Exercise Behaviors and Preferences questions, the means and standard deviations of related responses, and correlations between SPA and Exercise Behaviors and Preferences questions. The mean scores for each question were moderate, although the range of scores covered all possible responses for each question. The highest rated questions indicated, on average, a modest preference for attending aerobic dance class with a friend, participating in classes where most of the people are obviously in worse shape than the exerciser, and loose-fitting exercise attire. The lowest rated questions, on average, suggested weak dislikes for people spectating the aerobic dance class and wearing tight-fitting exercise clothing.

As can be observed in Table 5, 8 of the 12 Exercise Behaviors and Preferences questions were significantly correlated with SPA (both in simple correlations and partial correlations controlling for BF%), indicating identifiable patterns of response to questions in the context of consideration of SPA. Interestingly, no significant associations were found between SPA and a preference for attending class with a friend, the effects of being the fittest person in the class, or a liking for personal instructor feedback (3 of the 4 questions responded to most positively) nor with a preference for socializing in the aerobics class.

Table 5 Exercise Behaviors and Preferences Questions, Descriptive Statistics, and Correlations With SPA

Question	<i>M</i>	<i>SD</i>	Range	SPA corr.	SPA partial corr. (BF%)
1. To what extent do you prefer to aerobic dance in tight-fitting clothing?	2.55	1.16	1–5	–.39***	–.35***
2. To what extent do you prefer to aerobic dance in loose-fitting clothing?	3.52	1.08	1–5	.33***	.28**
3. When you are aerobic dancing, to what extent do you think about your appearance?	3.10	1.12	1–5	.39***	.43***
4. How does it affect you when most people in the aerobic dance class are obviously in better shape than you?	2.77	1.03	1–5	–.33***	–.32**
5. How does it affect you when most people in the aerobic dance class are obviously in worse shape than you?	3.54	0.91	1–5	.14	.12
6. How much do you think your body’s appearance is being judged by others while you are participating in an aerobic dance class?	3.02	0.97	1–5	.36***	.41***
7. How does it affect you when “spectators” (people not in the aerobics class) watch the class?	2.54	1.02	1–5	–.30**	–.23*
8. Is socializing in the aerobic class important to you?	2.70	1.18	1–5	.15	.16
9. To what extent do you prefer to attend aerobics class with a friend?	3.93	1.06	1–5	.17	.13
10. To what extent do you like to have personal feedback from the instructor while you are exercising?	3.33	1.15	1–5	.05	.04
11. To what extent do you like co-ed classes?	3.15	1.06	1–5	–.34***	–.32**
12. To what extent are you aware of the presence of members of the opposite sex in your aerobics class?	3.26	1.09	1–5	.33***	.40***

Note. SPA = Social physique anxiety.

* $p < .05$. ** $p < .01$. *** $p < .001$.

Discussion

The purpose of this investigation was to extend the Crawford and Eklund (1994) investigation of associations among SPA, REI, and ATEs by conducting the examination in the context of body composition. Interesting consistencies and contrasts with the previous investigation emerged, raising questions that require further investigation.

First, important consistencies with the reporting of Crawford and Eklund (1994) were observed with regard to associations between SPA and REI. Examination of descriptive statistics revealed REI scores to be largely consistent with the findings of the previous investigation. Further, significant positive correlations were observed between SPA and self-presentational reasons for exercise—motives related to development or maintenance of easily observable physical qualities (Body Tone, Physical Attractiveness, Weight Control). Also as observed in the previous investigation, reasons for exercise that were relatively independent of self-presentation (Fitness, Mood Enhancement, and Enjoyment) were unrelated to SPA, with the interesting exception of Health. Unlike Crawford and Eklund's (1994) previous findings, exercising for Health was found to be significantly and negatively associated with SPA, indicating that a greater emphasis upon exercising for health related reasons was associated with lower levels of SPA.

Hierarchical analysis to control for body composition in the examination of associations between SPA and REI also yielded results largely consistent with previous findings. Specifically, BF% was found to account for a significant amount of variance on the first step, but nonetheless, self-presentational reasons for exercise were still found to enhance prediction in the second step (although Weight Control was the only statistically significant contributor). Also consistent with the previous investigation, nonpresentational reasons for exercise contributed to the prediction of SPA in the third step (although exercising for Health was the only statistically significant contributor to the regression model). Crawford and Eklund (1994) had found Fitness to act as a suppressor variable in the previous investigation, and it appears that the nonpresentational reasons for exercise, as a group, acted in a similar fashion in this investigation.

Second, the associations between Exercise Behaviors and Preferences questions and SPA—although quite moderate, accounting for 9–15% of the variance in simple correlations—were largely consistent with what would be expected on the basis of the self-presentational perspective of Arkin (1981). Furthermore, the relationships between SPA and Exercise Behaviors and Preferences were found to be independent of body composition. Questions regarding a focus upon appearance and concerns about being judged while exercising were positively associated with SPA, whereas questions regarding amenability toward situations laden with evaluation potential (spectators and being the least fit class member) were negatively associated with SPA.

Interestingly, although no significant associations were observed between SPA and video presentations featuring attire emphasizing or de-emphasizing the physique in this investigation, significant associations were observed between questions probing preferences for working out in tight- or loose-fitting aerobic attire and SPA. Specifically, SPA was negatively associated with preferences for participating in tight-fitting aerobic dance clothing and was positively associated with preferences for loose-fitting aerobic clothing. The moderate ratings for

awareness of males in the aerobics class and for liking of coed classes indicated, on average, no particular like or dislike in this regard. However, the significant associations between these ratings and SPA indicated that women high in SPA tended to be less amenable to coed classes (at least in comparison to low-SPA peers) and more aware of the presence of members of the opposite sex attending classes. Thus, the emergence of systematic SPA-related patterns of responses to these questions suggest that self-presentation does contribute to exercise behaviors and preferences.

Although the desirability of socializing, attending aerobics classes with a friend, receiving personal feedback from the instructor, or being the most fit member of an aerobic dance class may be important from a self-presentational perspective (Hart & Gill, 1993), responses to those Exercise Behaviors and Preferences questions were not associated with SPA. Given that none of those questions inherently suggest potential for negative evaluation, perhaps it should have been expected that they would be unrelated to SPA as an index of self-presentational concerns related to the physique.

Crawford and Eklund (1994) previously found significant associations between SPA and ATEs, and as might be expected, the strongest association between SPA and ATEs was found relative to the Aerobics fashion attire video wherein the physique was particularly salient. However, in direct contrast, no associations were observed between SPA and ATEs in this investigation in any of the correlations or in the repeated-measures regression. Examination of appropriate scatterplots did not reveal any patterns implicating nonlinearity as contributory to this observation. This is an interesting contrast to both the findings of Crawford and Eklund (1994) and the previously mentioned associations between SPA and preferences for tight- or loose-fitting aerobic attire among the women in this investigation.

When compared to other investigations, it appears that the women in this investigation were not atypical in terms of SPA. Specifically, examination of descriptive statistics revealed that SPAS scores ($M = 37.31$, $SD = 9.81$) were marginally lower than those observed by Crawford and Eklund (1994; $M = 40.12$, $SD = 9.75$) but consistent with average values ($M = 37.9$, $SD = 9.78$, and $M = 37.0$, $SD = 10.1$) obtained across two samples of women by Hart et al. (1989). Importantly, the variability of SPAS scores in all samples appear to be virtually identical. By contrast, the results of this study did differ from Crawford and Eklund (1994) with regard to ATEs scores. In both investigations the shorts and T-shirt attire exercise setting was rated as more favorable than the aerobics fashion attire exercise setting; however, the mean comparative favorability score (discrepancy between ATEs shorts and T-shirts and aerobic fashion attire ratings) in this investigation was 2.19 ($SD = 4.24$), whereas Crawford and Eklund (1994) observed a substantially higher comparative favorability score of 5.38 ($SD = 6.72$). Given that SPA scores do not seem to be atypical in this investigation and that associations between SPA and other variables examined were consistent with self-presentational perspectives, it appears that this divergence may relate to the identified differences in ATEs responses. Nonetheless, it could be concluded that one of these contrasting results was anomalous or, alternatively, that there was some systematic difference(s) between the two investigations that led to the opposing outcomes.

The hypothesis that there may have been some systematic differences

between the two investigations that lead to the contrasting outcomes may be tenable. Specifically, although the procedures and materials for the two investigations were largely the same, there were two possibly significant exceptions. First, body composition was not assessed in the first investigation. Some have noted that the measurement of skinfolds may be a threatening process to some individuals (Hart et al., 1989; Riley, 1990), and hence, prior knowledge of collection of such data may have served as a factor deselecting potential volunteers uneasy with such procedures. Second, although Crawford and Eklund (1994) recruited subjects from a wide variety of liberal arts classes, the current investigation recruited subjects only from physical education major and activity classes. It is logical to expect that there is some sort of self-selection process involved in enrolling in these types of classes. Further, participation in such classes may have some influence on attitudes toward exercise settings via educational experience and/or desensitization to such settings.

Certain data available from the investigation suggest that it may be reasonable to implicate selection factors as a source for the contrasting results between studies. For example, it appears that the latter sample were more active than the women who volunteered for the previous investigation. Specifically, the women in the present sample averaged 6.5 hours per week of exercise and/or physical recreation, and further, virtually all indicated that they enjoyed exercise. By contrast, the women in Crawford and Eklund's (1994) initial investigation, although quite active, indicated that they worked out about 3.1 hours per week.

Furthermore, although the typical, almost unanimous, preference among collegiate women for a lower-than-actual weight was observed (Crawford & Eklund, 1994; Miller et al., 1980; Silberstein et al., 1988), the magnitude of the discrepancy appears to be less than in the previous investigation. Specifically, despite having similar heights (less than an inch different) and being slightly heavier (136.3 lb vs. 131.1 lb), women in this investigation, on average, indicated an ideal weight approximately 6% (8 lb) less than their current weight, whereas the previous investigation (Crawford & Eklund, 1994) found an average discrepancy of approximately 9% (12 lb). Further, Miller et al. (1980) had previously found collegiate women reporting an average desired ideal weight 14 lb below current weight.

Finally, although comparisons of absolute body composition statistics must be made very cautiously (Pollock & Wilmore, 1990), it appears that the women in the present investigation may also differ in body composition from broader samples of women their age. For example, Behnke and Wilmore (1974) reported average body composition among 20- to 24-year-old women to be 27% body fat. Among the college women in Hart et al.'s (1989) validation studies, BF% averaged about 32% for high-SPA women and 28% for low-SPA women. The women in the present study (perhaps not unexpectedly, given activity and weight satisfaction levels) appear to have lower BF% than are typical for women that age, with an average of about 22% body fat.

Hence, it is likely that the decision to recruit women solely from physical education major and activity classes or to collect body composition data may have introduced a selection bias affecting results in this investigation. However, these results may raise interesting questions regarding the operation of self-presentation in exercise. More precisely, if one considers (a) the results of the previous investigation, (b) the substantial amount of individual subject variability

(unexplained by BF%, SPA, and WS) associated ATES in the repeated-measures regression in this investigation, and (c) the observation that this sample does not appear to be atypical with regard to SPA, one can reasonably conclude that it may be profitable to explore factors that may mediate or moderate the associations between SPA and attitudes toward exercise settings. Specifically, future research questions might include the following: What sort of factors may contribute to selective disinhibition regarding situations saturated with evaluative potential of the physique? Are there other dispositional factors that interact with SPA leading to this sort of observation? Can individuals be desensitized to self-presentational concerns in certain situations either through exposure or through educational experience?

Further, a number of interesting questions emerge from the observations that SPA was associated with reasons for exercise and with exercise behaviors and preferences, even when body composition was factored out. For example, what is the basis of SPA concerns if, in fact, they are not entirely due to body composition? To what degree can other physical characteristics that are socially defined as desirable be implicated (e.g., height, somatotype)? Also, what sort of self-referent cognition (e.g., self-efficacy) and affect (e.g., self-esteem, body esteem) can be implicated as sources of SPA? Does one's motivational orientation in assessing competence in such situations contribute to SPA? Although promising, there are many questions that bear examination in applying self-presentational perspectives to exercise.

In summary, the results of this investigation indicate that self-presentation may be a significant factor influencing exercise behaviors. Considerable consistency has emerged across investigations with regard to associations between SPA and self-presentational motives for exercise regardless of whether or not body composition was taken into account. Further, associations between SPA and questions regarding exercise behaviors and preferences were largely consistent with what would be expected on the basis of self-presentational theoretical perspectives. Rather than undermining the tenability of potential contribution of self-presentational perspectives for understanding exercise behavior, the inability to fully replicate Crawford and Eklund's (1994) findings has raised interesting questions with regard to variables that may moderate or mediate self-presentational anxiety in exercise settings.

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Acknowledgments

We would like to gratefully acknowledge the assistance of Kathleen Buysse, Jennifer Levi, and Lori Watkins, of Auburn University, in data collection and data entry; Dr. David Pascoe, also of Auburn University, for his direction regarding the measurement of body composition; Dr. Lise Gauvin, of Concordia University, and Dr. Bob Harris, of the University of North Dakota, for their contributions to the statistical analysis; and Dr. James Whitehead, of the University of North Dakota, for his thought-provoking comments during manuscript preparation.

Manuscript submitted: May 12, 1994

Revision received: August 20, 1994

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